

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

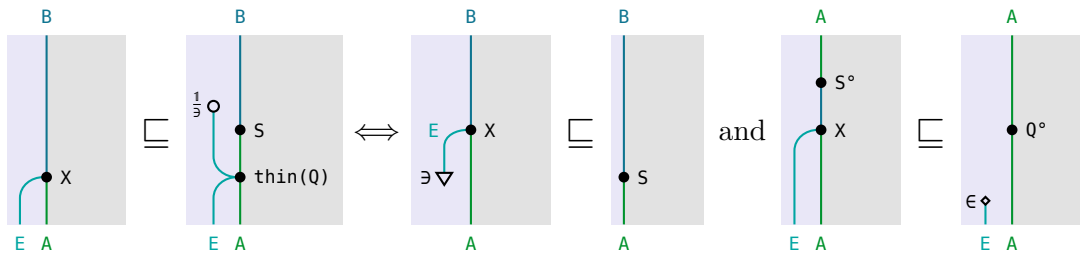
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

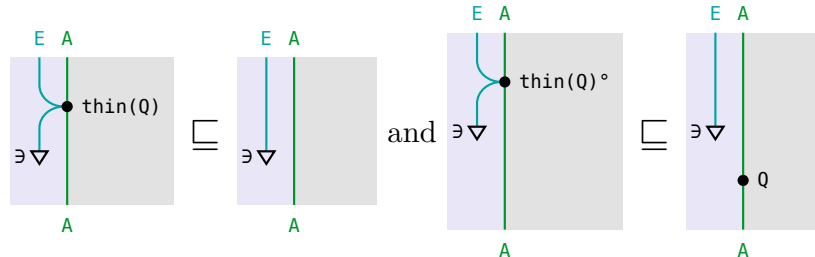
the law	what it says
$X \in \frac{S}{\exists} \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \epsilon \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \epsilon \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \in R \implies \text{thin}(Q) \in \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \in Q \implies 1 \in \text{thin}(Q)$ $Q \in Q \implies \text{thin}(Q) \text{ thin}(Q) \in \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \in R \wedge 1 \in Q \wedge R \in R \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ $Q \in R, 1 \in Q, R \in R$ — (8.1.2a); weaker than the book's “both preorders”: Q transitive is never used and $1 \in R$ follows	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{ thin}(Q) \supseteq \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \in Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{ union} \in \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \in (S^\circ \exists) \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \in \text{thin}(Q)$, since $\exists \circ \exists = \wedge(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ given $Q \in R, 1 \in Q, R \in R$

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$ (8.1.2a)
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) $1 \sqsubseteq \text{thin}(Q)$ — (8.1.1a)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$ (8.1.2b)
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) $\text{est}(R) \sqsubseteq \exists$ — est-laws
(b) $\text{thin}(Q) \exists \sqsubseteq \exists$ — (8.1.1a)

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (8.1.2c)
the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (8.1.1a)
(b) $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws
(c) R transitive

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$ (8.2)
the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$
every member of the singleton is a member of the set

(a) $\triangleq \text{thin}, \exists/\exists$ half
(b) $\frac{1}{\exists} \exists = 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \exists$ — est-laws

$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$
the singleton's member stands in Q to every member of the set

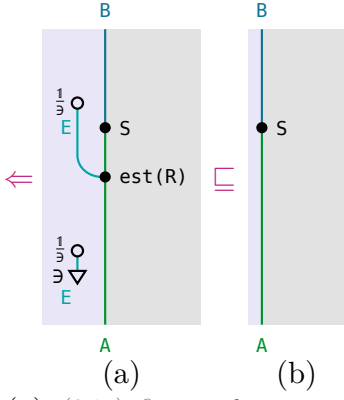
(a) $\triangleq \text{thin}, \in \setminus (Q^\circ \in)$ half
(b) $\in \text{est}(Q) \sqsubseteq Q^\circ$ — est-up
(c) $\frac{1}{\exists} \sqsubseteq \in$ — pow-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)}$$

(8.3)

given $\mathbf{Rn}(S^\circ S) \sqsubseteq Q$, Q a preorder

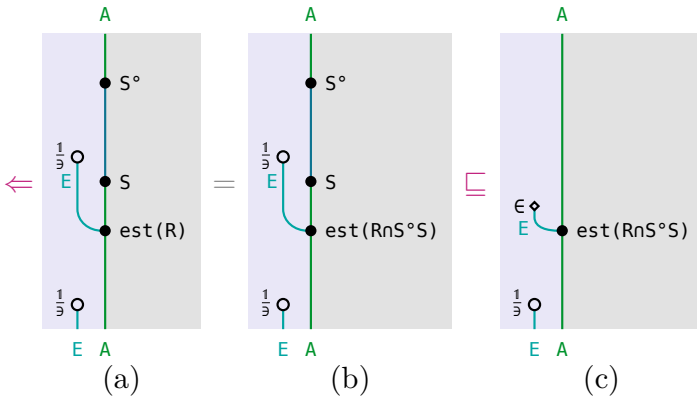
$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \exists \sqsubseteq S$$

every member of the singleton is a value S returns

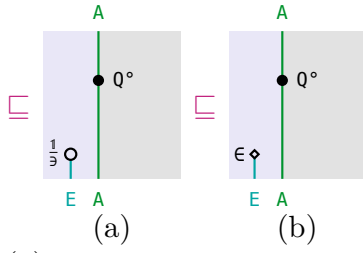
(a) (8.1a), first condition

(b) $\frac{1}{\exists} \exists = 1$ — pow-laws; $\frac{S}{\exists} \text{est}(R) \sqsubseteq S$ — est-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument

(a) (8.1a), second condition

(b) $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(RnS^\circ S)$ — est-laws(c) $S^\circ \frac{S}{\exists} \sqsubseteq E$ — pow-laws(a) $\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$ — est-up; $\mathbf{Rn}(S^\circ S) \sqsubseteq Q$ (b) $\frac{1}{\exists} \sqsubseteq E$ — pow-laws

Theorem 8.1

$$QQ \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

(8.1a) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle (S) \rangle$ for its S

(a) fusion, hylomorphism theorem
 (b) $S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — mon-str, relator-laws
 (c) $S^\circ F(\epsilon) \frac{F(\exists)S}{\exists} \sqsubseteq \epsilon$ — pow-laws
 (d) $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$ — $\triangleq \text{thin}$, adj-all

Corollary 8.1

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge RR \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$$

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders

(a) thinning theorem
 (b) $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (8.1a)

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.
 $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.
 $\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.
 $Q \triangleq Rn(\text{head head}^\circ)$, $S \triangleq F(1, \exists)\alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

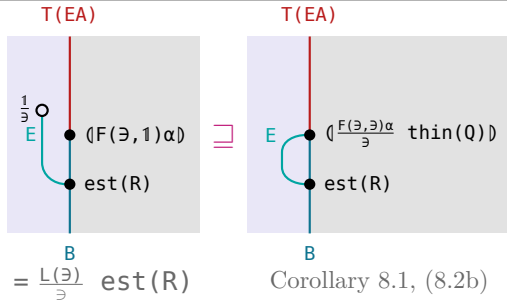
(8.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $Rn(S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \quad (8.2c)$$

$$\text{est}(R) \sqsubseteq \text{est}(R) \sqsubseteq \frac{F(\exists, 1)\alpha}{\exists} \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



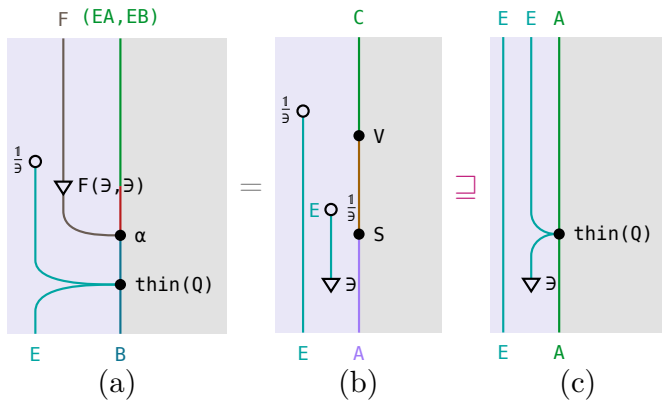
$\sqsubseteq$ (8.2d) under $\downarrow$ $\sqsupset$ monotonic: the whole chain runs inside the reduce, and the **est**(R) behind it never moves.

$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (8.2c)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

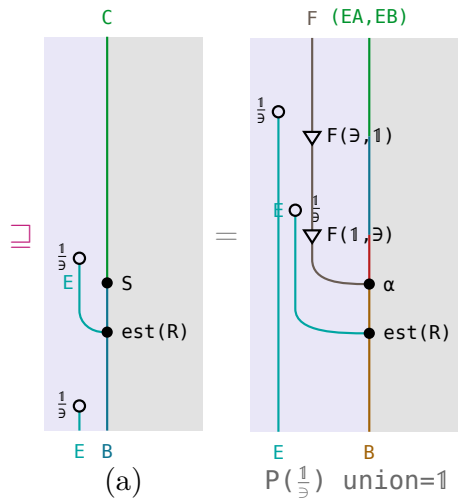
$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



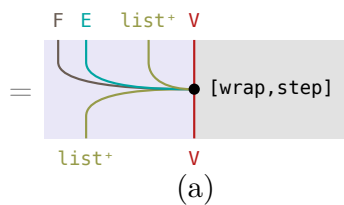
$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases [wrap, step]



$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{ — } \triangle \text{ cost}$$

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\llbracket \text{nil}, \text{bump } Q \rrbracket$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \sqsubseteq a \rightarrow [a] \# xs, a \sqsubseteq b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

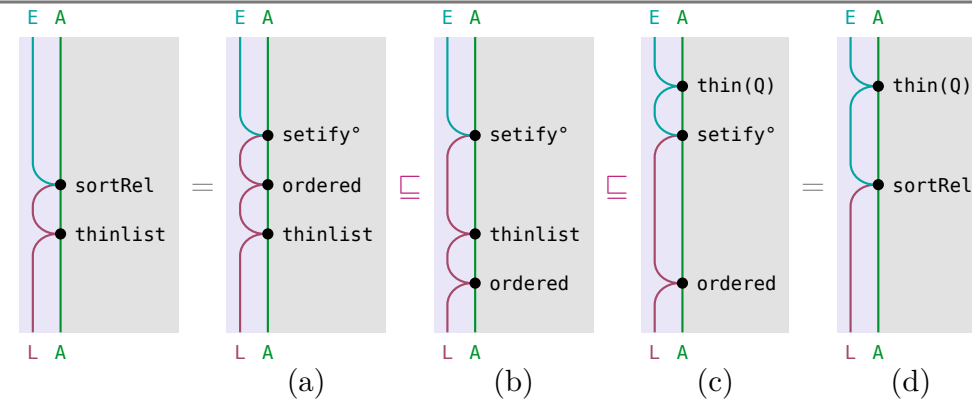
(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered subseq $\sqsubseteq$ subseq ordered $\wedge$ thinlist (8.6)

setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sortRel thinlist $\sqsubseteq$ thin(Q) sortRel

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.

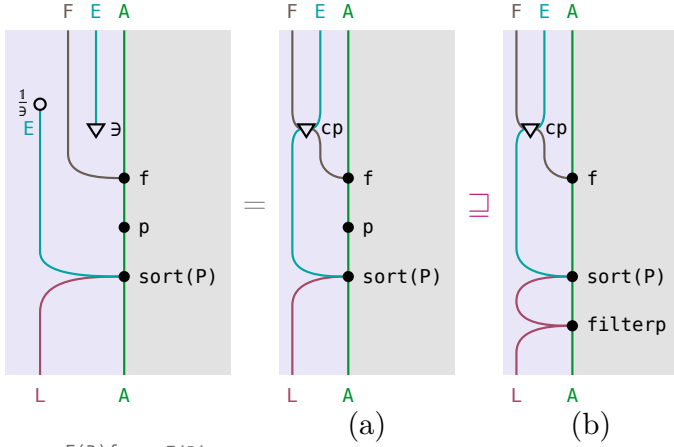


- (a) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (b) $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (c) $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$, dom-laws, triple-chains
 (d) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

Lemma 8.1

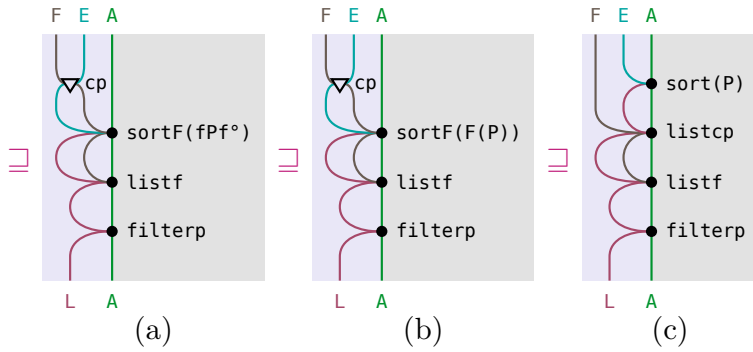
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge$
 $\text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$
 $\text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic
 on P , p coreflexive, F linear.

$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp } P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



- (a) $\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist
 $E(f) = P(f)$, f a map — powrel-laws
 (b) $\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P)$ — (8.3b)

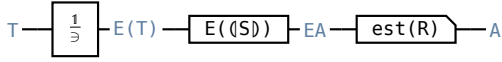
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$
 $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp } P(f) \text{sort}(P) \text{filterp}$
 combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



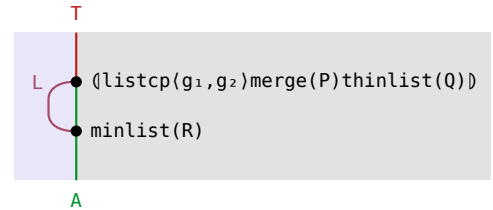
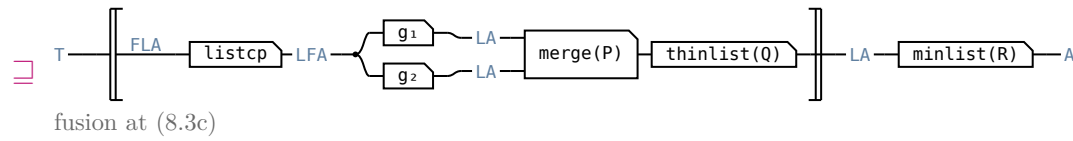
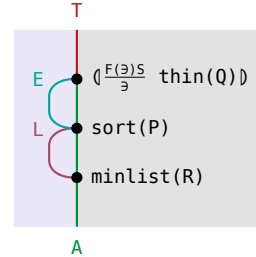
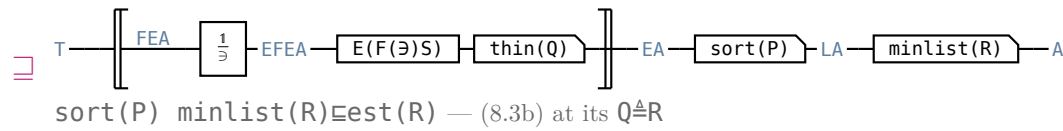
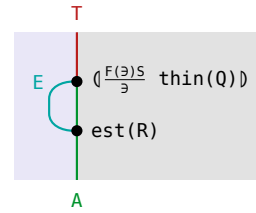
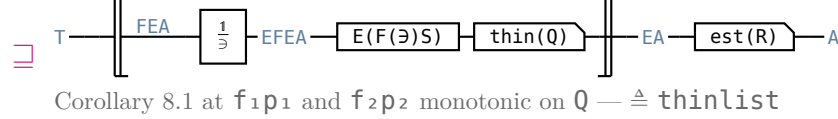
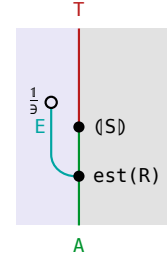
- (a) $\text{sort}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P)$ — (8.3b)
 (b) $FP \sqsubseteq fP f^\circ$ — mon-str, $\triangleq$ thinlist
 (c) $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP)$ — (8.3b)

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \subseteq R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge \text{MonoAlg}(f_2 \text{ } p_2, Q)$ Theorem 8.2
 $\wedge (\forall X. \forall Y. X \in Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ } P f_1^\circ) \text{listf}_1 \subseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ } P f_2^\circ) \text{listf}_2 \subseteq P(f_2) \text{sort}(P) \wedge$
 $\text{sort}(P) \text{filterp}_1 \subseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \subseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \subseteq \text{cp}$
 $\text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup } \text{sort}(P) \wedge$
 $\text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \subseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filterp}_1 \wedge$
 $g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp}(g_1, g_2) \text{merge}(P) \text{thinlist}(Q))$
 $\text{minlist}(R) \subseteq \frac{(S)}{S} \text{est}(R)$
 a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit



Hinze–Marsden



§8.4. The knapsack problem

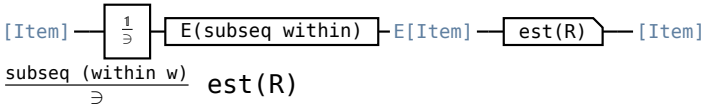
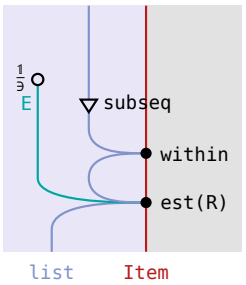
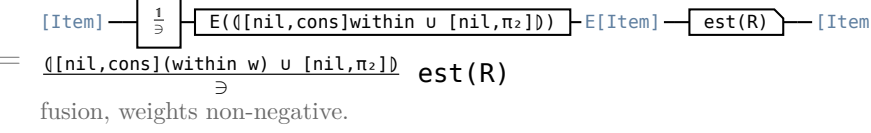
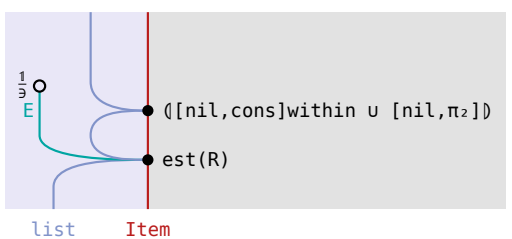
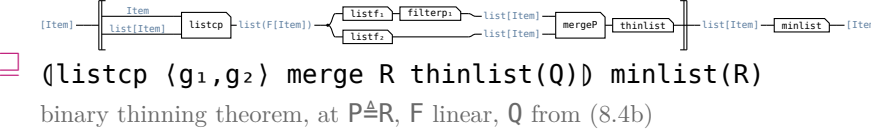
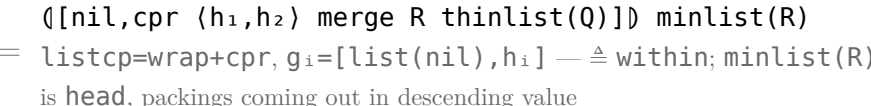
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(8.4b)

the law	what it says
$(1 \times R) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \quad (8.4c)$$

$$\begin{aligned} & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\ & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP} \text{filterp}_1 \sqsubseteq E(\text{within}) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \text{mergeP} \\ & \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq within}}{\supset} \text{est}(R) \\ & \text{the knapsack problem, as a fold that thins the packings kept at each item} \end{aligned}$$

circuit	Hinze–Marsden
	
	
	
	

§8.5. The paragraph problem

(8.5a)

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Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$ $\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ, .$	both halves are monotonic on Q once ties in waste are broken by the first line
merge $T = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(8.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([wrap \ wrap, new] T [wrap \ wrap, new]^\circ) \text{listf}_1 \sqsubseteq P([wrap \ wrap, new]) \text{sortP} \wedge \text{sortF}([wrap \ wrap, glue] T [wrap \ wrap, glue]^\circ) \text{listf}_2 \sqsubseteq P([wrap \ wrap, glue]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cpr}$
 $\text{sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sortP} \wedge \text{sortP minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{minlist} \sqsubseteq \frac{\text{partition fits}}{3} \text{est}(R))$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{3} \text{est}(R)$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([wrap \ wrap, new] u [wrap \ wrap, glue] ok)} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([wrap \ wrap, new] u ([wrap \ wrap, glue] (ok \ w)))}{3} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([wrap \ wrap, new] u ([wrap \ wrap, glue] (ok \ w)))} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([wrap \ wrap, new] u ([wrap \ wrap, glue] (ok \ w)))}{3} \text{est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}$.	
$\text{listcp} \langle g_1, g_2 \rangle \text{ cat thinlist}(Q) \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq T$ with merge $T = \text{cat}$, Q from (8.5b)	
$([start, cpr \langle h_1, h_2 \rangle \text{ cat thinlist}(Q)] \text{ minlist}(R)$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct — } \triangleq \text{partition}$	

$FA = (City \times City) + (City \times A)$, the base functor of cons-lists of length at least two; $listcp = wrap + cpr$; $tc : City \times City \rightarrow Real$, neither positive nor symmetric.

$$\text{cost}(x_s, y_s) = \text{outcost } x_s + \text{incost } y_s, \quad \text{outcost}[a_0, \dots, a_n] = \text{tc}(a_0, a_1) + \dots + \text{tc}(a_{n-1}, a_n), \quad \text{incost}[a_0, \dots, a_n] = \text{tc}(a_1, a_0) + \dots + \text{tc}(a_n, a_{n-1}).$$
$$\text{next} \triangleq \text{tail } \text{head}, \text{ next2} \triangleq \text{next} \times \text{next}, \text{ head2} \triangleq \text{head} \times \text{head}, R \triangleq \text{cost} \leq \text{cost}^\circ, Q \triangleq Rn(\text{next2 } \text{next2}^\circ)n(\text{head2 } \text{head2}^\circ), P \triangleq T, \\ g_1 \triangleq \text{list}([\text{start}, \text{dropl}]), g_2 \triangleq \text{list}([\text{start}, \text{dropr}]).$$

the law	what it says
$(1 \times R) \text{ drop}l \sqsubseteq \text{drop}l \text{ } R$ FALSE $(1 \times R) \text{ drop}r \sqsubseteq \text{drop}r \text{ } R$ FALSE	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists
$(1 \times Q) \text{ drop}l \sqsubseteq \text{drop}l \text{ } Q$ $(1 \times Q) \text{ drop}r \sqsubseteq \text{drop}r \text{ } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
& \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\
& \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp} \\
& \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \text{thinlist} \sqsubseteq \text{thin}(\text{Qc}(\text{tc})) \text{sortP} \wedge \text{sortP} \\
& \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{est}(\text{R}) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

circuit

Hinze–Marsden

